

Hodge Atoms as Occurrence-Indexed Marker Ledgers

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Abstract

Hodge atoms are defined from connected components of the reduced unramified Euler spectral cover of a maximal non-archimedean A-model F -bundle, followed by elementary identifications supplied by disjoint unions, blowups, and projective bundles. We isolate the categorical mechanism behind this construction. Expanding the degree of each spectral component into labelled occurrences, we form the thin groupoid generated by the elementary correspondences and the free commutative monoid on its connected components. Its universal fold recovers the Hodge-atom chemical formula, makes the three operation laws formal, and separates the abstract atom quotient from the generally coarser quotient by isomorphism of geometric atomic F -bundles. Killing the generators carried in dimension at most $d-2$ gives an effective quotient in which weak factorization makes the ledger birationally invariant for smooth projective d -folds. As a one-step consequence, if an atom of a smooth n -fold cannot be carried in dimension at most $n-1$, then every rank-two projective bundle carrying that occurrence is irrational. All quantum-cohomological comparison results enter only through an explicit provider record.

Keywords. Hodge atoms; quantum cohomology; marker ledger; weak factorization; projective bundle; irrationality.

1 The mechanism before the abstraction

Suppose that a smooth projective variety X determines finitely many local blocks a , each with a positive multiplicity $m_X(a)$. A comparison theorem for a blowup does more than equate two unlabelled multisets: it matches the blocks of the blowup with the ambient blocks and with a prescribed number of copies of the center blocks. Likewise, a projective-bundle comparison matches its blocks with copies of the base blocks. The elementary operation laws become formal once each multiplicity is expanded into separate occurrences and these comparison bijections are retained.

There are three successive constructions:

$$\text{local blocks with occurrences} \longrightarrow \text{thin comparison groupoid} \longrightarrow \text{free effective monoid.} \quad (1.1)$$

The first arrow remembers exactly which occurrences the provider identifies. The second forgets everything except their equivalence classes and multiplicities. A numerical marker is then simply a homomorphism out of the free monoid. Thus comparison is required before folding, but equality of richer block objects after folding is not.

Katzarkov–Kontsevich–Pantev–Yu construct Hodge atoms from the Euler spectral cover of a maximal non-archimedean A-model F -bundle [4, Sections 5.2–5.4]. Their multiplicities

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are degrees of connected components of the unramified reduced cover. Their blowup and projective-bundle decomposition theorems provide exactly the occurrence correspondences used in (1.1). The purpose of this note is to make the resulting specialization and its universal property explicit.

The distinction between occurrences and isomorphism classes is essential. It prevents a multiplicity from disappearing when two blocks are isomorphic, and it separates two quotients that need not agree: the abstract atom quotient generated by elementary geometric correspondences, and the quotient of geometric atomic F -bundles by isomorphism. The rationality criterion uses the former.

Here is the theorem ladder. Theorem 2.2 constructs the universal effective ledger from any occurrence provider. Theorem 3.1 identifies the standard Hodge-atom chemical formula with its Hodge specialization. Theorem 2.3 kills precisely the low-dimensional generators contributed by weak-factorization centers. Corollary 4.1 is the rank-two, one-step consequence.

The numerical direct-QDM atom used for cubic one-step irrationality in [5] is a sibling specialization of the same ledger mechanism, not a Hodge atom under another name. It uses a smaller block type and a different fold. No identification between the two carrier groupoids is needed here.

2 The occurrence-indexed ledger

Fix a universe of smooth projective varieties over a characteristic-zero field and choose a skeleton. This harmless choice only avoids set-theoretic duplication; isomorphisms remain among the generating correspondences.

Definition 2.1 (Occurrence provider). An occurrence provider assigns to every smooth projective X a finite set A_X and a function $m_X : A_X \rightarrow \mathbf{N}_{>0}$. Its expanded occurrence set is

$$\text{Occ}(X) := \{(a, j) : a \in A_X, 1 \leq j \leq m_X(a)\}. \quad (2.1)$$

The provider includes bijections, compatible with isomorphisms, of the following forms:

$$\text{Occ}(X \sqcup Y) \simeq \text{Occ}(X) \sqcup \text{Occ}(Y), \quad (2.2)$$

$$\text{Occ}(\text{Bl}_Z X) \simeq \text{Occ}(X) \sqcup (\text{Occ}(Z) \times \{1, \dots, c-1\}), \quad (2.3)$$

$$\text{Occ}(\mathbf{P}_X(E)) \simeq \text{Occ}(X) \times \{0, \dots, r-1\}, \quad (2.4)$$

where $Z \subset X$ is smooth of codimension $c \geq 2$ and E has rank $r \geq 2$. The labels j have no structure: a provider bijection is used only up to permuting the occurrences above a fixed element of A_X .

Let $\mathcal{O}$ be the disjoint union of the sets $\text{Occ}(X)$. First identify (a, j) with (a, j') for all labels over the same local block. Then use isomorphisms and each provider bijection, in either direction, to generate an equivalence relation on $\mathcal{O}$. Let Π be the associated thin groupoid: its objects are $\mathcal{O}$, and there is a unique arrow $x \rightarrow y$ exactly when the two occurrences are equivalent. Put

$$H := \pi_0(\Pi), \quad \mathcal{L} := \text{Sym}^{\sqcup}(H) = \mathbf{N}^{(H)}. \quad (2.5)$$

Thus $\mathcal{L}$ is the free commutative monoid on the occurrence classes. Write $q : \mathcal{O} \rightarrow H$ for the quotient map and define

$$\mathcal{B}(X) := \sum_{x \in \text{Occ}(X)} [q(x)] \in \mathcal{L}. \quad (2.6)$$

The blowup correspondence gives the commutative square

$$\begin{array}{ccc} \text{Occ}(\text{Bl}_Z X) & \xrightarrow{\sim} & \text{Occ}(X) \sqcup (\text{Occ}(Z) \times \{1, \dots, c-1\}) \\ q \downarrow & & \downarrow q^\sqcup \\ H & \xlongequal{\quad} & H. \end{array} \quad (2.7)$$

It records descent on objects. It does not assert an isomorphism between unrelated representatives of a common class.

Theorem 2.2 (Universal marker ledger). *The assignment $\mathcal{B}$ satisfies*

$$\mathcal{B}(X \sqcup Y) = \mathcal{B}(X) + \mathcal{B}(Y), \quad (2.8)$$

$$\mathcal{B}(\text{Bl}_Z X) = \mathcal{B}(X) + (c-1)\mathcal{B}(Z), \quad (2.9)$$

$$\mathcal{B}(\mathbf{P}_X(E)) = r\mathcal{B}(X). \quad (2.10)$$

For every commutative monoid M , a function $w_0 : \mathcal{O} \rightarrow M$ that is constant on the generating correspondences factors uniquely as

$$\mathcal{O} \xrightarrow{q} H \xrightarrow{w} M.$$

The function w in turn extends uniquely to a monoid homomorphism $\bar{w} : \mathcal{L} \rightarrow M$, and $I_w(X) := \bar{w}(\mathcal{B}(X))$ satisfies the folded versions of (2.8)–(2.10).

Proof. The provider bijections match every occurrence, including multiplicity. Changing a copy index is one of the generating arrows, so the construction is independent of the temporary labelling. After applying q , summing the two sides of (2.2)–(2.4) gives (2.8)–(2.10). A function on $\mathcal{O}$ is constant on the generated equivalence relation precisely when it factors uniquely through H . The universal property of the free commutative monoid then gives the unique extension $\bar{w}$. Applying that homomorphism to the three ledger identities proves the last assertion. $\square$

This formulation makes multiplicity bookkeeping literal. If a component has degree m , it contributes m objects to $\text{Occ}(X)$ and hence m copies of one generator to $\mathcal{B}(X)$. No symmetric-algebra convention or passage to a Grothendieck group is needed.

2.1 The effective dimension quotient

For $e \geq 0$, let $H_{\leq e} \subset H$ be the classes represented by an occurrence on some smooth projective variety of dimension at most e . The projection

$$\rho_e : \mathcal{L} = \mathbf{N}^{(H)} \longrightarrow \mathcal{L}_{>e} := \mathbf{N}^{(H \setminus H_{\leq e})} \quad (2.11)$$

deletes the low-dimensional coordinates. Because both monoids are free and effective, $\rho_e(\ell) \neq 0$ exactly when ℓ has a positive coefficient on a class outside $H_{\leq e}$.

Theorem 2.3 (Weak-factorization quotient). *Let $d \geq 2$. The assignment*

$$X \longmapsto \rho_{d-2}(\mathcal{B}(X)) \quad (2.12)$$

is a birational invariant of smooth projective d -folds. In particular, if $\rho_{d-2}(\mathcal{B}(X)) \neq 0$, then X is irrational.

Proof. Projective weak factorization expresses a birational map between smooth projective d -folds as blowups and blowdowns along smooth centers of codimension at least two [1, Theorem 0.1.1]. Every center has dimension at most $d - 2$, so $\rho_{d-2}(\mathcal{B}(Z)) = 0$. Equation (2.9) therefore shows that (2.12) is unchanged at each step.

For projective space, (2.10) applied over a point gives $\mathcal{B}(\mathbf{P}^d) = (d + 1)\mathcal{B}(\text{Spec } k)$. The point classes lie in $H_{\leq d-2}$, since $d \geq 2$, so the quotient ledger of $\mathbf{P}^d$ is zero. A rational X has the same quotient ledger, proving the contrapositive. $\square$

3 The Hodge-atom specialization

We now apply the preceding construction to the standard Hodge atoms of Katzarkov–Kontsevich–Pantev–Yu. Only the data needed for the type map are recalled; the non-archimedean A-model comparison theorems remain geometric inputs.

For a smooth complex projective variety X , let B_X^{Hod} be the Hodge-fixed locus in the base of its maximal non-archimedean A-model F -bundle. On the open set $U_X \subset B_X^{\text{Hod}}$ where the reduced Euler spectral cover is unramified, write

$$\tilde{U}_X \longrightarrow U_X \quad (3.1)$$

for that finite cover. Its connected components are the local Hodge atoms, and the multiplicity of a component is its degree over U_X [4, Definition 5.10 and Section 5.4]. Accordingly, set

$$A_X := \pi_0(\tilde{U}_X), \quad m_X(a) := \deg(\tilde{U}_{X,a}/U_X). \quad (3.2)$$

Automorphisms of X act on this weighted set. In the construction of Section 2, their orbits are imposed by isomorphism correspondences rather than silently discarded.

The required provider record is as follows.

Operation	F -bundle decomposition	Occurrence consequence
Disjoint union	external direct sum	disjoint union of covers
Blowup in codimension c	ambient summand plus $c - 1$ center summands	(2.3)
Rank- r projective bundle	r base summands	(2.4)

The spectral decomposition is [4, Theorem 4.1]. The blowup row is [4, Theorem 4.5], based on Iritani’s blowup decomposition [2, Theorem 5.18]. The projective-bundle row is [4, Theorem 4.11], based on the Iritani–Koto decomposition [3, Theorem 5.1]. These theorems identify the relevant spectral covers over connected nonempty domains and preserve the degrees of their components. Expanding equal degrees supplies occurrence bijections; different choices differ only by the internal relabellings already imposed in Section 2.

Theorem 3.1 (Hodge atoms are a ledger specialization). *For the provider (3.2), the set $H = \pi_0(\Pi)$ of Section 2 is the abstract Hodge-atom set HAtoms of [4, Definition 5.16 and Section 5.4]. Under this identification,*

$$\mathcal{B}(X) = \sum_{\alpha \in \text{HAtoms}} m_\alpha(X)[\alpha] =: \text{CF}_{\text{Hdg}}(X) \quad (3.3)$$

is its Hodge-atom chemical formula, and $H_{\leq e} = \text{HAtoms}_{\dim \leq e}$.

Proof. The objects of Π are precisely the local components of (3.1), expanded by their degrees. After the internal occurrence labels are identified, its remaining generating arrows are the isomorphism, disjoint-union, blowup, and projective-bundle elementary correspondences used to define abstract atoms in [4, Sections 5.2.3–5.2.6]. Passing to connected components of the thin groupoid is therefore the same generated quotient. Summing its occurrences gives (3.3). Finally, both dimension filtrations ask whether the same abstract class has a representative on a smooth projective variety of dimension at most e . $\square$

Combining Theorems 3.1 and 2.3 recovers the Hodge-atom non-rationality criterion [4, Proposition 5.30]. The ledger proof additionally exhibits the universal fold: every commutative-monoid-valued statistic of Hodge-atom occurrences that is constant on the elementary correspondences factors uniquely through $\mathcal{L}$.

Remark 3.2 (Abstract and geometric atoms). The geometric atomic F -bundles of [4, Definition 5.21] form a different quotient. Their Proposition 5.22 gives a map from abstract atoms to geometric atomic F -bundle classes, but does not assert that it is injective. The groupoid Π , the chemical formula (3.3), and the dimension filtration use abstract classes. Consequently an invariant first defined on geometric F -bundle classes becomes an atom marker only after its invariance under every elementary correspondence has been proved.

4 The one-step carrier criterion

For $h \in H$, define its carrier height by

$$\text{ht}(h) := \min\{\dim W : h \text{ occurs in } \mathcal{B}(W), W \text{ smooth projective}\}. \quad (4.1)$$

Equivalently, $h \in H_{\leq e}$ if and only if $\text{ht}(h) \leq e$. This is the dimension of a smallest carrier, not the dimension of a spectral-cover component or of the base of an atomic F -bundle.

Corollary 4.1 (Rank-two one-step criterion). *Let Z be a smooth projective n -fold with $n \geq 1$, let $E \rightarrow Z$ be a vector bundle of rank two, and let h occur in $\mathcal{B}(Z)$. If*

$$\text{ht}(h) > n - 1, \quad (4.2)$$

then $\mathbf{P}_Z(E)$ is irrational. In particular, a top-height class $\text{ht}(h) = n$ obstructs rationality after this one projective step.

Proof. The total dimension is $d = n + 1 \geq 2$. By (2.10),

$$\mathcal{B}(\mathbf{P}_Z(E)) = 2\mathcal{B}(Z),$$

so the coefficient of h remains positive. Condition (4.2) says $h \notin H_{\leq d-2}$. Hence

$$\rho_{d-2}(\mathcal{B}(\mathbf{P}_Z(E))) \neq 0,$$

and Theorem 2.3 applies. $\square$

For Hodge atoms, Corollary 4.1 is the rank-two specialization of the standard carrier-filtration criterion. The role of the categorical construction is exact but modest: it proves that the chemical formula and every lawful marker are folds of one occurrence ledger. It does not prove that a particular Hodge atom has the required carrier height; that remains a geometric exclusion statement.

The relation with the direct-QDM specialization of [5] can now be stated without conflating their objects.

Datum	Hodge-atom specialization	Direct-QDM specialization
Local object	component of the reduced unramified Euler spectral cover	typed generic numerical small-QDM block
Generated relation	elementary geometric correspondences	regular block isomorphisms supplied by QDM comparison
Effective fold	chemical formula and dimension quotient	chosen numerical block marker
One-step input	carrier-height exclusion	low-dimensional marker nullity

Both columns instantiate Theorem 2.2. They do not use the same groupoid, and a class in one column is not automatically a class in the other. This typed distinction is what permits the companion to clarify the standard atom specialization without enlarging the proof of cubic one-step irrationality.

The general projective-bundle correspondence in Definition 2.1 is part of the standard Hodge-atom quotient and is also needed to evaluate projective space. The stabilization consequence of this note is only Corollary 4.1, for a rank-two bundle; no claim is made for two or more projective stabilization factors.

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